

1 Questions

Nick's Question 1.1
In the Wave equation paper you showed that for the Greens function, we have that

$$\mathcal{F}G(t, \cdot)(\xi) = \frac{\sin(t|\xi|)}{|\xi|}$$

and so we see that

$$\mathcal{F}G(t, \cdot)(c\xi) = \frac{1}{c} \mathcal{F}(ct, \cdot)(\xi)$$

However in the Fractional SPDE paper the greens function is more complicated and we have that

$$\begin{aligned} \mathcal{F}Y_{\alpha, \beta, \gamma, d}(t, \cdot)(\xi) &= t^{\beta+\gamma-1} E_{\beta, \beta+\gamma}(-2^{-1} \nu t^\beta |\xi|^\alpha) \\ &= t^{\beta+\gamma-1} H_{1,2}^{1,1} \left(2^{-1} \nu t^\beta |\xi|^\alpha \middle| \begin{matrix} (0, 1) \\ (0, 1) \end{matrix} \right. (1 - \beta - \gamma, \beta) \Big) \\ &= t^{\beta+\gamma-1} \frac{1}{2\pi i} \int_L \mathcal{H}_1^{1,1}(s) \left(2^{-1} \nu t^\beta |\xi|^\alpha \right)^{-s} ds \end{aligned}$$

And so we see that

$$\mathcal{F}Y_{\alpha, \beta, \gamma, d}(t, \cdot)(c\xi) = t^{\beta+\gamma-1} \frac{1}{2\pi i} \int_L \mathcal{H}_1^{1,1}(s) \left(2^{-1} \nu t^\beta |c\xi|^\alpha \right)^{-s} ds$$

However I am unsure how to handle the constant, c , on the inside so we can get a nice scaling property like we do in the Wave Equation paper.

Also, instead of using the Fox-H function we can just say

$$\begin{aligned} \mathcal{F}Y_{\alpha, \beta, \gamma, d}(t, \cdot)(\xi) &= t^{\beta+\gamma-1} E_{\beta, \beta+\gamma}(-2^{-1} \nu t^\beta |\xi|^\alpha) \\ &= t^{\beta+\gamma-1} \sum_{k=0}^{\infty} \frac{(-2^{-1} \nu t^\beta |\xi|^\alpha)^k}{\Gamma(\beta k + (\beta + \gamma))} \end{aligned}$$

and so

$$\mathcal{F}Y_{\alpha, \beta, \gamma, d}(t, \cdot)(c\xi) = t^{\beta+\gamma-1} \sum_{k=0}^{\infty} \frac{(-2^{-1} \nu t^\beta |c\xi|^\alpha)^k}{\Gamma(\beta k + (\beta + \gamma))}$$

but I still don't see how to handle the c .

Here is how one should proceed:

$$\begin{aligned} \mathcal{F}Y_{\alpha, \beta, \gamma, d}(t, \cdot)(c\xi) &= t^{\beta+\gamma-1} \sum_{k=0}^{\infty} \frac{(-2^{-1} \nu t^\beta |c\xi|^\alpha)^k}{\Gamma(\beta k + (\beta + \gamma))} \\ &= t^{\beta+\gamma-1} \sum_{k=0}^{\infty} \frac{(-2^{-1} \nu t^\beta c^\alpha |\xi|^\alpha)^k}{\Gamma(\beta k + (\beta + \gamma))} \\ &= t^{\beta+\gamma-1} \sum_{k=0}^{\infty} \frac{(-2^{-1} \nu (tc^{\frac{\alpha}{\beta}})^\beta |\xi|^\alpha)^k}{\Gamma(\beta k + (\beta + \gamma))} \\ &= c^{\frac{\alpha}{\beta}(1-\gamma-\beta)} (tc^{\frac{\alpha}{\beta}})^{\beta+\gamma-1} \sum_{k=0}^{\infty} \frac{(-2^{-1} \nu (tc^{\frac{\alpha}{\beta}})^\beta |\xi|^\alpha)^k}{\Gamma(\beta k + (\beta + \gamma))} \end{aligned}$$

$$= c^{\frac{\alpha}{\beta}(1-\gamma-\beta)} \mathcal{F}Y_{\alpha,\beta,\gamma,d} \left(c^{\frac{\alpha}{\beta}} t, \cdot \right) (c\xi)$$

Please check if this scaling reduces to heat and wave equations correctly.

Nick's Question 1.2 In the wave equation paper with Xia Chen you said that that fundamental solution to the SPDE

$$\begin{cases} \frac{\partial^2 u}{\partial t^2}(t, x) - \frac{1}{2} \Delta u(t, x) = \sqrt{\Theta} u(t, x) \dot{W}(x) & x \in \mathbb{R}^d, t > 0 \\ u(0, \cdot) = 1 \\ u_t(0, \cdot) = 0 \end{cases} \quad (1.1)$$

wspde

and the mild solution is defined to be

$$u(t, x) = 1 + \sqrt{\Theta} \int_0^t \left(\int_{\mathbb{R}^d} G(t-s, x-y) u(s, y) W(\delta y) \right) ds \quad (1.2)$$

weqs

but for general initial conditions for the wave equation we have the solution to be

$$\begin{aligned} u(t, x) = & \frac{\partial}{\partial t} \int_{\mathbb{R}^d} G(t-s, x-y) u(0, y) dy + \int_{\mathbb{R}^d} G(t-s, x-y) u_t(0, y) dy \\ & + \sqrt{\Theta} \int_0^t \left(\int_{\mathbb{R}^d} G(t-s, x-y) u(s, y) W(\delta y) \right) ds \end{aligned} \quad (1.3)$$

gweqs

But when the initial conditions are such as in 1.1 then the first two integral in 1.3 are 1 and 0 respectively and so we arrive at 1.2.

In the fractional SPDE

$$\begin{cases} (\partial^\beta + \frac{\nu}{2}(-\Delta)^{\alpha/2}) u(t, x) = I_t^\gamma \left[\rho(u(t, x)) \dot{W}(x) \right] & x \in \mathbb{R}^d, t > 0 \\ u(0, \cdot) = \mu & \beta \in (0, 1] \\ u(0, \cdot) = \mu_0, \quad u_t(0, \cdot) = \mu_1 & \beta \in (1, 2] \end{cases} \quad (1.4)$$

fspde

and we define

$$J_0(t, x) = \begin{cases} (Z(t, \cdot) * \mu)(x) & \beta \in (0, 1] \\ (Z^*(t, \cdot) * \mu_0)(x) + (Z(t, \cdot) * \mu_1)(x) & \beta \in (1, 2] \end{cases}$$

and

$$I(t, x) = \iint_{[0,t] \times \mathbb{R}} Y(t-s, x-y) \rho(u(s, y)) W(\delta y) ds$$

and the general solution for 1.4 is

$$u(t, x) = J_0(t, x) + I(t, x) \quad (1.5)$$

feqs

My question is that if we consider the following fractional spde

$$\begin{cases} (\partial^\beta + \frac{\nu}{2}(-\Delta)^{\alpha/2}) u(t, x) = I_t^\gamma \left[\rho(u(t, x)) \dot{W}(x) \right] & x \in \mathbb{R}^d, t > 0 \\ u(0, \cdot) = 1 & \beta \in (0, 1] \\ u(0, \cdot) = 1, \quad u_t(0, \cdot) = 0 & \beta \in (1, 2] \end{cases} \quad (1.6)$$

fspde1

then can we calculate $J_0(t, x)$? More specifically is $J_0(t, x) = 1$ in this case? The wave equation paper worked out nicely because in that paper $J_0(t, x) = 1$ and so the solution was able to be written as 1.2. The 1 allows us to easily do the Piccard iteration to get equation (2.3) from the Wave equation paper with Xia Chen.

So if our solution can not be written as

$$u(t, x) = 1 + I(t, x)$$

I am not sure at the moment how we can come up with an expression for the f'_n s in the Weiner Expansion like you did in equation (2.3) in the paper with Xia Chen.

Hi, Nick, here is the answer:

Notice that, by Theorem 4.1 of my paper,

$$\begin{aligned} (Z_{\alpha, \beta, \gamma, d}(t, \cdot) * 1)(x) &= \mathcal{F}Z_{\alpha, \beta, \gamma, d}(t, \cdot)(0) \\ &= t^{\lceil \beta \rceil - 1} E_{\beta, \lceil \beta \rceil}(0) \\ &= \frac{t^{\lceil \beta \rceil - 1}}{\Gamma(\lceil \beta \rceil)}. \end{aligned}$$

Hence, when $\beta \in (0, 1]$, we see that $(Z_{\alpha, \beta, \gamma, d}(t, \cdot) * 1)(x) = 1$. When $\beta \in (1, 2]$,

$$\begin{aligned} (Z_{\alpha, \beta, \gamma, d}^*(t, \cdot) * 1)(x) &= \mathcal{F}Z_{\alpha, \beta, \gamma, d}^*(t, \cdot)(0) \\ &= E_{\beta}(0) \\ &= \frac{1}{\Gamma(1)} = 1. \end{aligned}$$

Hence, under the initial conditions given by (1.6), $J_0(t, x) \equiv 1$.

Nick's Question 1.3 Guess for the constant (1.20) from the Wave Equation Paper In the wave equation paper, the constant (1.20) was

$$C = \begin{cases} \frac{4 - \Delta}{3 - \Delta} & \text{under Assumption A} \\ \frac{4 - d}{3 - d} & \text{under Assumption B or C} \end{cases} \quad (1.7) \quad \boxed{\text{const}}$$

I believe in the fractional paper the constant will be

$$\begin{cases} \frac{2(\beta + \gamma - 1) + 2 - \beta\Delta/\alpha}{2(\beta + \gamma - 1) + 1 - \beta\Delta/\alpha} & \text{under Assumption A} \\ \frac{2(\beta + \gamma - 1) + 2 - \beta d/\alpha}{2(\beta + \gamma - 1) + 1 - \beta d/\alpha} & \text{under Assumption B or C} \end{cases}$$

I think this because in the wave equation paper where we use $(4 - \Delta)$ this gets replaced by $(2(\beta + \gamma - 1) + 2 - \beta\Delta/\alpha)$ and thus we can guess that

$$\begin{cases} 4 - \Delta & \rightarrow & 2(\beta + \gamma - 1) + 2 - \beta\Delta/\alpha \\ 3 - \Delta & \rightarrow & 2(\beta + \gamma - 1) + 1 - \beta\Delta/\alpha \end{cases}$$

Note that when $\alpha = \beta = 2$ and when $\gamma = 0$ then the guess reduces to the wave equation.

Nick's Question 1.4 Lemma 2.3 for the fractional equation

1. In the outline you sent me, you use

$$\mathcal{F}_a = \left\{ g \in H^a(\mathbb{R}^d) : \|g\|_{L^2(\mathbb{R}^d)} = 1, \mathcal{E}_a(g, g) < \infty \right\}$$

Where $H^a(\mathbb{R}^d)$ is the fractional Sobelov Space where $H^2(\mathbb{R}^d) = W^{1,2}(\mathbb{R}^d)$. However, I saw on line that $H^a(\mathbb{R}^d) = W^{a,2}(\mathbb{R}^d)$ and so $H^2(\mathbb{R}^d) = W^{2,2}(\mathbb{R}^d)$. Is there a mistake here?

2. Where is $\mathcal{M}$ defined with the sup over $\mathcal{F}_d$ and not $\mathcal{F}_a$? Is this a typo?
3. When defining $\mathcal{E}_a(g, g)$ why does the $(2\pi)^{-d}$ appear here but not in the wave paper?
4. How do we generalize ρ from the wave paper to the fractional equation? For the wave equation we have that

$$\rho = (\mathcal{M}(\gamma, 2))^{(4-\alpha)/2}$$

but why does it generalize to this in the fractional sense

$$\rho = (\mathcal{M}_a(\gamma, 2))^{(2a-\alpha)/a} = 2^{-a/\alpha} (\mathcal{M}_a(\gamma, 1))^{(2a-\alpha)/a}$$

which is what you put in the outline you sent me.

Nick's Question 1.5 : Typos from Xia Chen Paper (Large deviations for Riesz potentials of additive processes)

From what I can tell there seem to be several typos that I can't justify from this paper that seem serious. The following are some definitions and a Theorem from Xia Chen's Paper.

Definitions:

$$\rho = \sup_{\|f\|_2=1} \int_{\mathbb{R}^d} \left[\int_{\mathbb{R}^d} \frac{f(s+t)f(t)}{\sqrt{1+|s+t|^\beta} \sqrt{1+|t|^\beta}} dt \right] \varphi_{d-\alpha} ds \quad (1.8)$$

$$\mathcal{E}_\beta(f, f) = (2\pi)^{-d} \int_{\mathbb{R}^d} |s|^\beta |\hat{f}(s)|^2 ds \quad (1.9)$$

$$\mathcal{F}_\beta = \left\{ f \in L^2(\mathbb{R}^d) : \|f\|_2 = 1, \mathcal{E}_\beta(f, f) < \infty \right\} \quad (1.10)$$

$$\Lambda_\alpha = \sup_{f \in \mathcal{F}_\beta} \left\{ \left(\int_{\mathbb{R}^d} \frac{\prod_{j=1}^p f^2(x_j)}{|x_1 + \dots + x_p|^\alpha} \prod_{j=1}^p dx_j \right)^{1/p} - \mathcal{E}_\beta(f, f) \right\} \quad (1.11)$$

Theorem 1.5

$$\rho = (2\pi)^{-d} (\Lambda_\alpha)^{p-(\alpha/\beta)}$$

This theorem is proved in section 7 of the paper and starts with two lemmas.

Lemma 7.1: Lemma 7.1 claims that Λ_α is finite, however I believe that there is a typo in the proof. In the very first line of the proof, it references equation (A.2) and (A.2) is

proved by referencing (A.8). I believe that equation (A.8) is wrong. (A.8) says that

$$\int_{\mathbb{R}^{dp}} \frac{\prod_{j=1}^p f_j(x_j)}{|x_1 + \dots + x_p|^{d-\alpha}} \prod_{j=1}^p dx_j = \int_{\mathbb{R}^{d2}} \frac{f_1(x)(f_2 * \dots * f_p)(y)}{|x - y|^{d-\alpha}} dx dy$$

but if we let $p = 2$ then (A.8) says that

$$\int_{\mathbb{R}^{2d}} \frac{f_1(x_1)f_2(x_2)}{|x_1 + x_2|^{d-\alpha}} dx_1 dx_2 = \int_{\mathbb{R}^{d2}} \frac{f_1(x)f_2(y)}{|x - y|^{d-\alpha}} dx dy$$

which is not true.

Lemma 7.2

From first glance I think Lemma 7.2 is okay.

Proof of Theorem 1.5: First some definitions for this theorem.

Definitions:

1.

$$L(f) = \left(\int_{\mathbb{R}^{dp}} \frac{\prod_{j=1}^p |h(x_j)|^2}{|x_1 + \dots + x_p|^\sigma} \prod_{j=1}^p dx_j \right)^{1/p}$$

2.

$$\tilde{L}(f) = \mathcal{E}_\beta(f, f)$$

3.

$$J = \sup_{\|f\|_2^2 + \tilde{f}=1} L(f) = \sup_{f \in \mathcal{H}} \frac{L(f)}{\|f\|_2^2 + \tilde{L}(f)}$$

There are several places in the proof that I think are typos. Also I do not understand for sure why we have the equality in the supremums in definition (3) above.

1. First is equation (7.21) which says by Parseval's Relation we get

$$\sup_{\langle g, Qg \rangle=1} \int_{\mathbb{R}^d} [(Qg) * (\widetilde{Qg})]^p(-s) \varphi(s) ds = \sup_{\langle g, Qg \rangle=1} (2\pi)^{-d} \int_{\mathbb{R}^d} \mathcal{F}([(Qg) * (\widetilde{Qg})]^p)(x) \mathcal{F}\varphi(x) dx \quad (1.12)$$

typo1

where $\tilde{p}(x) = p(-x)$ and $g(x) = f(x)\sqrt{1+|x|^\beta}$ where f is the same function from the definition of ρ . Notice the left hand side of 1.12 has a $(-s)$ as the input of the function. Then when taking the Fourier transform of $[(Qg) * (\widetilde{Qg})]^p(-s)$ we should have the following

$$\mathcal{F}[(Qg) * (\widetilde{Qg})]^p(-x)$$

which follows from the property that if we let $\tilde{p}(x) = p(-x)$ then $\mathcal{F}(\tilde{p})(\xi) = \mathcal{F}(p)(-\xi)$.

2. Im going to continue, assuming that the previous typo was actually not a typo. The next typo is the unlabeled centered equation right above (7.22) which says

$$\begin{aligned} \sup_{\langle g, Qg \rangle=1} (2\pi)^{-d} \int_{\mathbb{R}^d} \mathcal{F}([(Qg) * (\widetilde{Qg})]^p)(x) \mathcal{F}\varphi(x) dx \\ = \sup_{\langle g, Qg \rangle=1} (2\pi)^{-d(p+1)} \int_{\mathbb{R}^d} [|\widetilde{Qg}|^2]^{*p}(x) \mathcal{F}\varphi(x) dx \end{aligned} \quad (1.13)$$

typo2

It is stated in the paper that $\mathcal{F}(f * g) = \mathcal{F}(f)\mathcal{F}(g)$ and $\mathcal{F}(fg) = (2\pi)^{-d}\mathcal{F}(f) * \mathcal{F}(g)$ and f^{*p} is the p -fold convolution. So using these rules we should get that

$$\begin{aligned} & \sup_{\langle g, Qg \rangle = 1} (2\pi)^{-d} \int_{\mathbb{R}^d} \mathcal{F}\left([(Qg) * (\widetilde{Qg})]^p\right)(x) \mathcal{F}\varphi(x) dx \\ &= \sup_{\langle g, Qg \rangle = 1} (2\pi)^{-d} \int_{\mathbb{R}^d} (2\pi)^{-pd} \mathcal{F}\left([(Qg) * (\widetilde{Qg})]\right)^{*p}(x) \mathcal{F}\varphi(x) dx \\ &= \sup_{\langle g, Qg \rangle = 1} (2\pi)^{-d(1+p)} \int_{\mathbb{R}^d} \left[\widehat{Qg} \widehat{\widetilde{Qg}} \right]^{*p}(x) \mathcal{F}\varphi(x) dx \end{aligned}$$

It seems as if they are assuming that $\widehat{\widetilde{Qg}} = \widehat{Qg}$ which is only true if Qg is an even function.

3. **Details for (7.22)** Again I am going to continue as if the previous was not a typo. Im giving the details for (7.22) just for clarity. (7.22) says that if we define $h = (2\pi)^{-d}\widehat{Qg}$ then

$$\begin{aligned} & \sup_{\langle g, Qg \rangle = 1} (2\pi)^{-d(p+1)} \int_{\mathbb{R}^d} [|\widehat{Qg}|^2]^{*p}(x) \frac{1}{|x|^\alpha} dx \\ &= \sup_{(2\pi)^d \|h\|_2^2 + (2\pi)^d \tilde{L}(h) = 1} (2\pi)^{-d(p-1)} \int_{\mathbb{R}^d} [|h|^2]^{*p}(x) \frac{1}{|x|^\alpha} dx \end{aligned}$$

Since $(2\pi)^d h = \widehat{Qg}$ then this implies that $\hat{h} = \widetilde{Qg} = Q\tilde{g}$ since Q is even and because when considering $\hat{f}(x) = \int \exp(ixs)f(s)ds$ as the definition of the Fourier transform then we have the property that $\hat{\hat{f}} = (2\pi)^d \tilde{f}$. Also we have that

$$\langle g, Qg \rangle = \langle \tilde{g}, \widetilde{Qg} \rangle = \langle \hat{h}Q^{-1}, \hat{h} \rangle = (2\pi)^d \|h\|_2^2 + (2\pi)^d \tilde{L}(h)$$

since for this definition of the Fourier transform we have $\langle f, g \rangle = (2\pi)^{-d} \langle \hat{f}, \hat{g} \rangle$. Putting this together gives us (7.22).

4. **Equation (7.23)** I am unsure why a change of variable implies that

$$\begin{aligned} & \sup_{(2\pi)^d \|h\|_2^2 + (2\pi)^d \tilde{L}(h) = 1} (2\pi)^{-d(p-1)} \int_{\mathbb{R}^d} [|h|^2]^{*p}(x) \frac{1}{|x|^\alpha} dx \\ &= \sup_{\|h\|_2^2 + \tilde{L}(h) = 1} (2\pi)^{-d} \int_{\mathbb{R}^{dp}} \frac{\prod_{j=1}^p |h(x_j)|^2}{|x_1 + \dots + x_p|^\sigma} \prod_{j=1}^p dx_j \end{aligned}$$

In particular how the sup limits changed.

Possible Justification: The integrals can be shown to be equal through a change of variables.

$$\begin{aligned} & \int_{\mathbb{R}^d} [|h|^2]^{*p}(x) \frac{1}{|x|^\alpha} dx = \int_{\mathbb{R}^d} (h * \dots * h)(x) \frac{1}{|x|^\alpha} dx \\ &= \int_{\mathbb{R}^d} \int_{\mathbb{R}^{d(p-1)}} h(x - y_1) h(y_1 - y_2) \dots h(y_{p-2} - y_{p-1}) h(y_{p-1}) \prod_{j=1}^{p-1} dy_j \frac{1}{|x|^\alpha} dx \end{aligned}$$

then applying the substitution

$$\begin{pmatrix} x - y_1 = u_1 \\ y_1 - y_2 = u_2 \\ \dots \\ y_{p-2} - y_{p-1} = u_{p-1} \\ y_{p-1} = u_p \end{pmatrix}$$

will transform the integral as

$$\int_{\mathbb{R}^d} (h * \dots * h)(x) \frac{1}{|x|^\alpha} dx = \int_{R^{dp}} \frac{\prod_{j=1}^p |h(x_j)|^2}{|x_1 + \dots + x_p|^\sigma} \prod_{j=1}^p dx_j$$

As for the limits of the supremum, note that

$$(2\pi)^d \|h\|_2^2 + (2\pi)^d \tilde{L}(h) = 1 \implies \|(2\pi)^{d/2} h\|_2^2 + \tilde{L}((2\pi)^{d/2} h) = 1$$

Therefore using the above definitions of J , $L(f)$ and $\tilde{L}(f)$ we see that

$$\begin{aligned} & \sup_{(2\pi)^d \|h\|_2^2 + (2\pi)^d \tilde{L}(h) = 1} (2\pi)^{-d(p-1)} \int_{R^{dp}} \frac{\prod_{j=1}^p |h(x_j)|^2}{|x_1 + \dots + x_p|^\sigma} \prod_{j=1}^p dx_j \\ &= \sup_{\|(2\pi)^{d/2} h\|_2^2 + \tilde{L}((2\pi)^{d/2} h) = 1} (2\pi)^{-d(p-1)} L(h)^p \\ &= (2\pi)^{-d(p-1)} \left[\sup_{\|(2\pi)^{d/2} h\|_2^2 + \tilde{L}((2\pi)^{d/2} h) = 1} L(h) \right]^p \\ &= (2\pi)^{-d(p-1)} \left[\sup_{h \in \mathcal{H}} \frac{L(h)}{\|(2\pi)^{d/2} h\|_2^2 + \tilde{L}((2\pi)^{d/2} h)} \right]^p \\ &= (2\pi)^{-d(p-1)} \left[(2\pi)^d \sup_{h \in \mathcal{H}} \frac{L(h)}{\|h\|_2^2 + \tilde{L}(h)} \right]^p \\ &= (2\pi)^{-d} J^p \end{aligned}$$

which is exactly equation (7.23) in the Xia Chen paper.

2 Proof Sketches

Nick's Question 2.1 A mild solution for

$$\begin{cases} (\partial^\beta + \frac{\nu}{2}(-\Delta)^{\alpha/2}) u(t, x) = I_t^\gamma [\rho(u(t, x)) \dot{W}(x)] & x \in \mathbb{R}^d, t > 0 \\ u(0, \cdot) = 1 & \beta \in (0, 1] \\ u(0, \cdot) = 1, \quad u_t(0, \cdot) = 0 & \beta \in (1, 2] \end{cases}$$

must satisfy the following integral equation

$$u(t, x) = 1 + \int_0^t \left(\int_{\mathbb{R}^d} u(t-s, x-y) Y(s, y) W(\delta y) \right) ds$$

an through a Piccard Iteration Scheme we can see that the solution should be given by

$$u(t, x) = 1 + \sum_{k=1}^{\infty} \theta^{k/2} I_k(f_k(\cdot, x, t)) \quad (2.1) \quad \text{sol}$$

where

$$f_k(x_1, \dots, x_k, x, t) = \int_0^t \int_0^{t_k} \cdots \int_0^{t_2} Y(t - t_k, x - x_k) \cdots Y(t_2 - t_1, x_2 - x_1) dt_1 \cdots dt_n \quad (2.2) \quad \text{solker}$$

$$= \int_0^t \int_0^{t_k} \cdots \int_0^{t_2} Y(t_1, x - x_k) \cdots Y(t_k - t_{k-1}, x_2 - x_1) dt_1 \cdots dt_n \quad (2.3)$$

and also

$$\mathcal{F}f_k(\cdot, x, t)(\xi_1, \dots, \xi_n) = e^{-i(\sum_{j=1}^n \xi_j) \cdot x} \int_{[0, t]^n} \prod_{k=1}^n \overline{\mathcal{F}Y(t_{t_k+1} - t_k, \cdot)} \left(\sum_{j=1}^k \xi_j \right) dt \quad (2.4) \quad \text{fsolker}$$

where $t_{n+1} = t$

Nick's Question 2.2 Greens Function Scaling Property

For and $c > 0$ we have

$$\mathcal{F}Y(ct, \cdot)(\xi) = c^{b+r-1} \mathcal{F}Y(t, \cdot)(t^{b/a} \xi). \quad (2.5) \quad \text{fscale}$$

which is shown above.

Nick's Question 2.3 $\mathcal{H}^{\otimes n}$ Norm Scaling Property

Suppose that $\mu(d\xi) = \varphi(\xi)d\xi$ and φ satisfies $\varphi(c\xi) = c^{-(d-\alpha)}$ for some $\alpha \in (0, d)$ and for any $c > 0$. Also recall that

$$\mathcal{F}Y(ct, \cdot)(\xi) = c^{b+r-1} \mathcal{F}Y(t, \cdot)(t^{b/a} \xi).$$

Then

$$\left\| \tilde{f}_n(\cdot, 0, t) \right\|_{H^{\otimes n}}^2 = t^{[2(b+r)-b\alpha/a]n} \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{H^{\otimes n}}^2 \quad (2.6) \quad \text{nscale}$$

Proof:

$$\left\| \tilde{f}_n(\cdot, 0, t) \right\|_{H^{\otimes n}}^2 = \frac{1}{n!} \sum_{\rho \in \Sigma_n} \int_{\mathbb{R}^{nd}} \mathcal{F}f_n(\cdot, 0, t)^2(\vec{\xi}) \varphi(\vec{\xi}) d\vec{\xi}$$

where $\varphi(\vec{\xi}) = \varphi(\xi_1) \cdots \varphi(\xi_n)$

$$\begin{aligned} \int_{\mathbb{R}^{nd}} \mathcal{F}f_n(\cdot, 0, t)^2 \varphi(\vec{\xi}) d\vec{\xi} &= \int_{\mathbb{R}^{nd}} \left(\int_{[0, t]^n} \overline{\mathcal{F}Y(t - t_n, \cdot)(\xi_1 + \cdots + \xi_n)} \cdots \overline{\mathcal{F}Y(t_2 - t_1, \cdot)(\xi_1)} \right)^2 \varphi(\vec{\xi}) d\vec{\xi} \\ &= t^{2n(b+r-1)} \int_{\mathbb{R}^{nd}} \left(\int_{[0, 1]^n} \overline{\mathcal{F}Y\left(1 - \frac{t_n}{t}, \cdot\right)(t^{b/a}(\xi_1 + \cdots + \xi_n))} \cdots \overline{\mathcal{F}Y\left(\frac{t_2}{t} - \frac{t_1}{t}, \cdot\right)(t^{b/a}\xi_1)} \right)^2 \varphi(\vec{\xi}) d\vec{\xi} \\ &= t^{2n(b+r-1)+2n} \int_{\mathbb{R}^{nd}} \left(\int_{[0, 1]^n} \overline{\mathcal{F}Y(1 - t_n, \cdot)(t^{b/a}(\xi_1 + \cdots + \xi_n))} \cdots \overline{\mathcal{F}Y(t_2 - t_1, \cdot)(t^{b/a}\xi_1)} \right)^2 \varphi(\vec{\xi}) d\vec{\xi} \end{aligned}$$

$$\begin{aligned}
&= t^{2n(b+r-1)+2n} \int_{\mathbb{R}^{nd}} \left(\int_{[0,1]^{n<}} \overline{\mathcal{F}Y(1-t_n, \cdot)(\xi_1 + \dots + \xi_n) \dots \mathcal{F}Y(t_2 - t_1, \cdot)(\xi_1)} \right)^2 \varphi(t^{-b/a} \vec{\xi}) t^{-ndb/a} d\vec{\xi} \\
&= t^{2n(b+r-1)+2n-nb\alpha/a} \int_{\mathbb{R}^{nd}} \left(\int_{[0,1]^{n<}} \overline{\mathcal{F}Y(1-t_n, \cdot)(\xi_1 + \dots + \xi_n) \dots \mathcal{F}Y(t_2 - t_1, \cdot)(\xi_1)} \right)^2 \varphi(\vec{\xi}) d\vec{\xi} \\
&= t^{2n(b+r-1)+2n-nb\alpha/a} \int_{\mathbb{R}^{nd}} \mathcal{F}f_n(\cdot, 0, 1)(\vec{\xi})^2 \varphi(\vec{\xi}) d\vec{\xi} \\
&= t^{[2(b+r-1)+2-b\alpha/a]n} \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{H^{\otimes n}}^2 \\
&= t^{[2(b+r)-b\alpha/a]n} \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{H^{\otimes n}}^2
\end{aligned}$$

Nick's Question 2.4 Lemma 2.3 We define

$$\mathcal{F}_a = \left\{ g \in H^a(\mathbb{R}^d) : \|g\|_{L^2(\mathbb{R}^d)} = 1, \mathcal{E}_a(g, g) < \infty \right\}$$

and then we also define

$$\mathcal{M}_a(f, \theta) = \sup_{g \in \mathcal{F}_a} \left\{ \langle g^2 * f, g^2 \rangle^{1/2} - \frac{\theta}{2} \int_{\mathbb{R}^d} |\xi|^a |\mathcal{F}g(\xi)|^2 d\xi \right\} \quad (2.7) \quad \boxed{\text{var}}$$

Now we define $\tilde{g}(x) = B^{-d/2}g(x/B)$. Its easy to show that $\mathcal{F}\tilde{g}(\xi) = B^{d/2}\mathcal{F}g(B\xi)$. By a change of variables, we can then show that

$$\int_{\mathbb{R}^d} |\xi|^a |\mathcal{F}\tilde{g}(\xi)|^2 d\xi = B^{-a} \int_{\mathbb{R}^d} |\xi|^a |\mathcal{F}g(\xi)|^2 d\xi$$

Next, by applying the scaling property of f , $f(cx) = c^{-\alpha}f(x)$, we can show by again using a change of variables that

$$\begin{aligned}
(\tilde{g}^2 * f)(x) &= B^{-d} \int_{\mathbb{R}^d} \tilde{g}^2(y/B) f(x-y) dy \\
&= \int_{\mathbb{R}^d} g^2(u) f(x-uB) du \\
&= B^{-\alpha} \int_{\mathbb{R}^d} g^2(u) f\left(\frac{x}{B} - u\right) du \\
&= B^{-\alpha} (g^2 * f)(x/B)
\end{aligned}$$

This can be used to show that

$$\langle \tilde{g}^2 * f, \tilde{g}^2 \rangle = B^{-\alpha} \langle g^2 * f, g^2 \rangle$$

Now we can rewrite (2.7) as the following

$$\mathcal{M}_a(\Theta f, \theta) = \frac{1}{\theta} \sup_{g \in \mathcal{F}_a} \left\{ \frac{\sqrt{\Theta}}{\theta} B^{\alpha/2} \langle \tilde{g}^2 * f, \tilde{g}^2 \rangle^{1/2} - \frac{B^a}{2} \int_{\mathbb{R}^d} |\xi|^a |\mathcal{F}\tilde{g}(\xi)|^2 d\xi \right\}$$

and as before, in order to preserve a scaling property, we want to choose B such that

$$\frac{\sqrt{\Theta}}{\theta} B^{\alpha/2} = B^a$$

and solving for B we get

$$B = \left(\frac{\sqrt{\Theta}}{\theta} \right)^{\frac{2}{2a-\alpha}}$$

and after plugging this B in we get that

$$\mathcal{M}_a(\Theta f, \theta) = \Theta^{\frac{a}{2a-\alpha}} \theta^{-\frac{\alpha}{2a-\alpha}} \mathcal{M}_a(f, 1)$$

Some definitions for solvability (details in my notebook)

$$\begin{aligned} L_n(y_1, \dots, y_n) &= n! \int_0^\infty e^{-t} \tilde{f}_n(y_1, \dots, y_n, 0, t) dt \\ &= \sum_{\sigma \in \Sigma_n} \int_0^\infty e^{-t} \int_{[0,t]^n} \prod_{k=1}^n Y(s_k - s_{k-1}, y_{\sigma(k)} - y_{\sigma(k-1)}) ds dt \end{aligned}$$

where $s_0 = y_{\sigma(0)} = 0$.

$$\begin{aligned} H_n(t, \mathbf{x}) &= n! \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n K(x_k - y_k) \tilde{f}_n(y_1, \dots, y_n, 0, t) d\mathbf{y} \\ &= \sum_{\sigma \in \Sigma_n} \int_{[0,t]^n} \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n K(x_k - y_k) \prod_{k=1}^n Y(s_k - s_{k-1}, y_{\sigma(k)} - y_{\sigma(k-1)}) d\mathbf{y} ds \end{aligned}$$

$$L(x) = \int_0^\infty e^{-t} Y(t, x) dt \quad \text{and} \quad \mathcal{F}L(\xi) = \int_0^\infty e^{-t} \mathcal{F}Y(t, \cdot)(\xi) dt = \frac{1}{1 + \frac{\nu}{2} |\xi|^a}$$

$$L_n(y_1, \dots, y_n) = \sum_{\sigma \in \Sigma_n} L(y_{\sigma(1)}) L(y_{\sigma(2)} - y_{\sigma(1)}) \cdots L(y_{\sigma(n)} - y_{\sigma(n-1)})$$

$$\mathcal{F}L_n(\xi_1, \dots, \xi_n) = \sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} \left| \sum_{j=k}^n \xi_j \right|^a}$$

$$T_n = \int_{(\mathbb{R}^d)^n} \left[\sum_{\sigma \in \Sigma_n} \prod_{k=1}^n \frac{1}{1 + \frac{\nu}{2} \left| \sum_{j=k}^n \xi_j \right|^a} \right]^2 \mu(d\xi_1) \cdots \mu(d\xi_n)$$

$$C'_\mu = \int_{(\mathbb{R}^d)^n} \left(\frac{1}{1 + |\xi|^a} \right)^2 \mu(d\xi)$$

Nick's Question 2.5 Lemma 3.3, 3.4, and 3.5 Same proof as in wave paper. Have all details in my notebook.

Nick's Question 2.6 Lemma 3.6 We have that

$$C'_\mu < \infty \quad \text{and} \quad T_n < (n!)^2 (C'_\mu)^n$$

proof: By the same spherical substitution in the wave paper we see that

$$C'_\mu = \left(\int_{S^{d-1}} \varphi(\gamma) d\sigma(\gamma) \right) \left(\int_0^\infty \frac{\rho^{\alpha-1}}{(1+\rho^a)^2} d\rho \right)$$

The first integral is shown to be finite in the wave paper. The second integral is finite if

$$2a - \alpha + 1 > 1 \tag{2.8} \quad \boxed{\text{alpha2}}$$

Second inequality has the same proof as in the wave paper however I do not know why

$$(n!)^2 \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n \left(\frac{1}{1 + |\sum_{j=k}^n \xi_j|^2} \right)^2 \mu(d\xi_1) \cdots \mu(d\xi_n) \leq (n!)^2 \left[\sup_{\eta \in \mathbb{R}^d} \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n \left(\frac{1}{1 + |\xi + \eta|^2} \right)^2 \mu(d\xi) \right]^n$$

Nick's Question 2.7 Lemma 3.7

$$\int_0^\infty e^{-t} \left\| \tilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^n dt \leq \frac{2^{n[2(b+r) - \frac{b\alpha}{a}]} T_n}{(n!)^2} \leq 2^{n[2(b+r) - \frac{b\alpha}{a}]} (C'_\mu)^n$$

Proof: Details in my notebook

Nick's Question 2.8 Theorem 3.1 1.6 has a unique solution $\{u(t, x) : t > 0, x \in \mathbb{R}^d\}$ and

$$\sup_{(t,x) \in [0,T] \times \mathbb{R}^d} \mathbb{E}(|u(t, x)|^p) < \infty \quad \forall p \geq 2 \text{ and } T > 0.$$

Proof: First using 2.5 (details in notebook) we can show that

$$\mathcal{F} \tilde{f}_n(\cdot, 0, ct)(\xi_1, \dots, \xi_n) = c^{n(b+r)} \mathcal{F} \tilde{f}_n(\cdot, 0, t)(c^{b/a} \xi_1, \dots, c^{b/a} \xi_n). \tag{2.9}$$

In order to prove the theorem we need to show that $\sum_{n \geq 0} \theta^n n! \left\| \tilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 < \infty$. From the scaling property 2.6 we see that

$$\begin{aligned} \int_0^\infty e^{-t} \left\| \tilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 dt &= \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \int_0^\infty e^{-t} t^{[2(b+r) - \frac{b\alpha}{a}]n} dt \\ &= \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \Gamma \left(\left[2(b+r) - \frac{b\alpha}{a} \right] n + 1 \right) \end{aligned}$$

as long as we have

$$2(b+r) - \frac{b\alpha}{a} > 0 \tag{2.10} \quad \boxed{\text{alpha3}}$$

By using Lemma 3.7 we can deduce that

$$\left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 = \frac{\int_0^\infty e^{-t} \left\| \tilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 dt}{\Gamma \left([2(b+r) - \frac{b\alpha}{a}] n + 1 \right)}$$

$$\begin{aligned}
&\leq \frac{2^{n[2(b+r)-\frac{b\alpha}{a}]}}{\Gamma([2(b+r)-\frac{b\alpha}{a}]n+1)} T_n \\
&\leq \frac{\left(2^{[2(b+r)-\frac{b\alpha}{a}]} C'_\mu\right)^n}{\Gamma([2(b+r)-\frac{b\alpha}{a}]n+1)}
\end{aligned}$$

By the same reasoning as in the proof in the wave paper (details in my notebook) we can say that there exists a constant depending on α , k_α , such that

$$\left\|\tilde{f}_n(\cdot, 0, 1)\right\|_{\mathcal{H}^{\otimes n}}^2 \leq k_\alpha \frac{\left(2^{[2(b+r)-\frac{b\alpha}{a}]} C'_\mu\right)^n}{(n!)^{2(b+r)-b\alpha/a}}$$

Therefore,

$$\begin{aligned}
\sum_{n \geq 0} \theta^n n! \left\|\tilde{f}_n(\cdot, 0, t)\right\|_{\mathcal{H}^{\otimes n}}^2 &\leq \sum_{n \geq 0} \theta^n n! k_\alpha \frac{\left(2^{[2(b+r)-\frac{b\alpha}{a}]} C'_\mu\right)^n}{(n!)^{2(b+r)-b\alpha/a}} \\
&= \sum_{n \geq 0} \theta^n k_\alpha \frac{\left(2^{[2(b+r)-\frac{b\alpha}{a}]} C'_\mu\right)^n}{(n!)^{2(b+r)-1-b\alpha/a}}
\end{aligned}$$

and the last sum is finite when

$$2(b+r) - 1 - b\alpha/a > 0 \tag{2.11}$$

alpha4